

Boost Converter Power Stage Calculation

Given parameters:

- Input Voltage: $V_{\text{in}} = 12V$
- Output Voltage: $V_{\text{out}} = 24V$
- Output Power: $P_{\text{out}} = 120W$
- Switching Frequency: $F_s = 100KHz$
- Ripple Current Rate: 30%
- Ripple Voltage Rate: 1%

- Calculate Duty Cycle
- Calculate Output Current
- Calculate Output Resistance
- Calculate Ripple Current, Maximum and Minimum Ripple Current
- Calculate Ripple Voltage
- Calculate Capacitor
- Calculate Inductor
- Calculate Selection of Mosfet
- Calculate Selection of Diode